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Title of paper: “NeuFair: Neural Network Fairness Repair with Dropout” by Vishnu Asutosh Dasu, Saeid Tizpaz-Niari et al.

Summary:

This paper introduces NeuFair, a novel post-processing method that improves the fairness of pre-trained deep neural networks by removing specific neurons during inference. The authors argue that some neurons encode biased representations and disproportionately contribute to unfair treatment of protected groups (e.g., race, sex). Instead of retraining models or modifying datasets, NeuFair looks for neuron dropout patterns that minimize unfairness (measured primarily by Equalized Odds Difference) while preserving the model's accuracy in predicting.

To efficiently explore the huge combinatorial space of possible dropout configurations, the research paper applies two randomized search algorithms: Simulated Annealing (SA) and Random Walk (RW). NeuFair is tested against seven DNN benchmarks from five socially sensitive domains, delivering up to a 69% decrease in unfairness with little or no utility deterioration. The research also examines hyperparameters, explains why randomized search is beneficial, and demonstrates experimentally that NeuFair outperforms other cutting-edge post-processing approaches such as DiCE.

Two points that you like about the paper (strengths):

- The concept of employing inference-time dropout as a fairness mitigation technique is novel, straightforward, and extremely practical, particularly in real-life scenarios where retraining (or even accessing) the original model or data is not viable.
- The authors conduct extensive experiments on several datasets, compare algorithms (SA, RW, and DiCE), examine hyperparameters, give diagnostic insights, and even explain why randomized algorithms can uncover fairness-improving states in high-dimensional spaces. Going into the “why” made me understand the concept a lot better.

Two points that you did not like about the paper (weaknesses):

- The research paper's approach is only assessed on one fairness measure (EOD) and binary classification tasks, limiting its applicability to other fairness concepts (e.g., calibration, statistical parity) or multi-class scenarios.
- While Simulated Annealing (SA) works well empirically, it does not provide theoretical assurances of optimality, and performance is heavily dependent on hyperparameters, time budget, and the search space. This can produce results that are inconsistent across datasets, which is problematic and does not make the method as optimal as the paper shows it to be.

Detailed comments and questions:

- The paper assumes that some neurons encode discriminatory behavior but does not deeply analyze which neurons are causing unfairness or how they develop such

representations. I think that a deeper, neuron-level interpretability study could strengthen this claim and better explain the behavior of the neurons.

- The largest DNNs utilized in the experiments appear relatively small by modern standards. Would NeuFair scale to networks with millions of neurons, as seen in contemporary NLP or vision models?
- Dropping neurons at inference may slightly alter computational patterns. In such cases, would NeuFair introduce extra latency in real-time deployment systems?
- Equalized Odds is stricter than other group fairness concepts. Would NeuFair act differently under statistical parity or equal opportunity constraints?